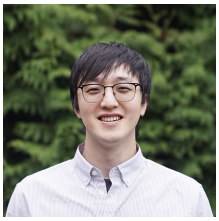
Jiayang Ren

Email: rjy12307@outlook.com
LinkedIn: linkedin.com/in/jiayang-ren
Google Scholar: scholar.google.com/citations?user=V3QBv3cAAAAJ



Academic Appointments

- **University of Wisconsin–Madison, Chemical and Biological Engineering** — Madison, USA
  Postdoctoral Researcher, Advisors: Dr. Styliani Avraamidou and Dr. Victor M. Zavala — *Oct 2026 – Present*

Education

- **University of British Columbia** — Vancouver, Canada
  Ph.D. in Chemical Engineering, Advisor: Dr. Yankai Cao, GPA: 98.0/100 — *Sep 2021 – May 2026*
- **Zhejiang University** — Hangzhou, China
  M.S. in Control Engineering, Advisor: Dr. Dong Ni, GPA: 88.9/100 — *Sep 2018 – Jun 2021*
- **Zhejiang University** — Hangzhou, China
  B.A. in Automation, GPA: 3.80/4.0 — *Sep 2014 – Jun 2018*

Research Interests

- Decision-focused and surrogate-based optimization.
- Global optimization algorithms for large-scale interpretable machine learning.
- Learning-based control with interpretable models.
- Data-driven modeling, monitoring, and control of manufacturing, energy, and process systems.

Journal Articles

- Zou, C., **Ren, J.**, Mao, Q., Liu, J., Lai, M., and Cao, Y. (2026). A Moving-Horizon Approximate Branch-and-Reduce Method for Deep Classification Trees. *Transactions on Machine Learning Research*. **J2C Certification**.
- Liu, J., Zou, C., **Ren, J.**, Fang, G., Li, C., Wang, Y., Ma, Z., and Hu, B. (2026). Continual Learning with Elastic Regularization and Synthetic Replay for Federated MLLM Fine-Tuning. *IEEE Journal of Selected Topics in Signal Processing*. Accepted.
- **Ren, J.**, Osuna-Enciso, V., and Cao, Y. (2026). A Moving-Horizon Differential Evolution Algorithm for Training Deep Classification Trees on Large Datasets. *Expert Systems with Applications*, 333(B): 133846.
- **Ren, J.**, You, N., Hua, K., Ji, C., and Cao, Y. (2026). A Global Optimization Algorithm for K-Center Clustering of One Billion Samples. *Management Science*, 72(6): 5106–5130.
- Guo, P., Song, Y., Wang, B., Cao, Y., **Ren, J.**, Ji, C., Liu, J., Zhang, Q., and Mao, Q. (2026). Class-Missing Semi-Supervised Document Key Information Extraction via Synergistic Refinement Estimation. *Information Processing & Management*, 63(1): 104335.
- Li, C.*, **Ren, J.***, Seth, A., Zhang, Y., Huang, J., Li, C., and Cao, Y. (2025). Hierarchical Model Predictive Control for Energy Consumption Regulation of Industrial-Scale Circulation Counter-Flow Paddy Drying Process. *Energy*, 321: 135431. (*co-first authors)
- Okamoto, M., **Ren, J.**, Mao, Q., Liu, J., and Cao, Y. (2024). Deep Learning-Based Approximation of Model Predictive Control Laws Using Mixture Networks. *IEEE Transactions on Automation Science and Engineering*, 22: 2909–2922.
- Li, Y., Wang, Y., Chen, Y., Lu, Y., Hua, K., **Ren, J.**, Mozafari, G., Lu, Q., and Cao, Y. (2022). Deep-Learning-Based Predictive Control of Battery Management for Frequency Regulation. *Industrial & Engineering Chemistry Research*, 61(24): 8432–8442.
- **Ren, J.**, and Ni, D. (2021). A Real-Time Monitoring Framework for Wafer Fabrication Processes with Run-to-Run Variations. *IEEE Transactions on Semiconductor Manufacturing*, 34(4): 483–492.
- **Ren, J.**, and Ni, D. (2020). A Batch-Wise LSTM-Encoder Decoder Network for Batch Process Monitoring. *Chemical Engineering Research and Design*, 164: 102–112.

Computer Science Conference Proceedings

- Mao, Q., **Ren, J.**, Wang, Y., Zou, C., Zheng, J., and Cao, Y. (2025). Differentiable Decision Tree via "ReLU+Argmin" Reformulation. *Advances in Neural Information Processing Systems (NeurIPS)*, 38: 166874–166909. **Spotlight** (top 3% of submissions).
- Liu, P., Hao, Z., Ren, X., Yuan, H., **Ren, J.**, and Ni, D. (2024). PAPM: A Physics-Aware Proxy Model for Process Systems. *41st International Conference on Machine Learning (ICML)*, PMLR 235: 31080–31105.
- **Ren, J.**, Hua, K., and Cao, Y. (2022). Global Optimal K-Medoids Clustering of One Million Samples. *Advances in Neural Information Processing Systems (NeurIPS)*, 35: 982–994.
- Hua, K., **Ren, J.**, and Cao, Y. (2022). A Scalable Deterministic Global Optimization Algorithm for Training Optimal Decision Tree. *Advances in Neural Information Processing Systems (NeurIPS)*, 35: 8347–8359.
- Shi, M., Hua, K., **Ren, J.**, and Cao, Y. (2022). Global Optimization of K-Center Clustering. *39th International Conference on Machine Learning (ICML)*, PMLR 162: 19956–19966.

Engineering Conference Proceedings

- **Ren, J.**, Juang, M.-H., Zhao, T., and Cao, Y. (2026). Fast, High-Performance, and Interpretable Quadcopter Control Using Decision Trees. *23rd IFAC World Congress*.
- Zhao, T., **Ren, J.**, Zou, C., Yang, Y., Cao, Y., and Gopaluni, R.B. (2026). Interpretable Reinforcement Learning for Multi-Loop Adaptive PID Control. *23rd IFAC World Congress*.
- **Ren, J.**, Mao, Q., Zhao, T., and Cao, Y. (2025). Exact Learning of Linear Model Predictive Control Laws Using Oblique Decision Trees with Linear Predictions. *IEEE Conference on Decision and Control (CDC)*, pp. 7018–7023.
- Wang, Y., Kumar, A., **Ren, J.**, You, P., Seth, A., Gopaluni, R.B., and Cao, Y. (2024). Interpretable Data-Driven Capacity Estimation of Lithium-Ion Batteries. *12th IFAC Symposium on Advanced Control of Chemical Processes (ADCHEM)*, IFAC-PapersOnLine 58(14): 139–144.
- **Ren, J.**, Hua, K., Trajano, H., and Cao, Y. (2023). Global Optimal Explainable Models for Biorefining. *33rd European Symposium on Computer Aided Process Engineering (ESCAPE)*, Computer Aided Chemical Engineering 52: 1339–1346.
- **Ren, J.**, and Ni, D. (2019). Real-Time Fault Detection System for Multiphase Plasma Etching Process Using OES, Two-Step Division and Change Stage Alignment Method. *2019 Chinese Automation Congress (CAC)*, pp. 599–604.

Software & Patents

- **Ren, J.**, Osuna-Enciso, V., and Cao, Y. (2026). MHDETrees.jl: Moving-Horizon Differential Evolution for Optimal Classification and Regression Trees. Open-source Julia package (MIT License), github.com/YankaiGroup/MHDETrees.jl.
- Ni, D., Zhu, F., and **Ren, J.** (2020). A Real-Time Measurement Method and Device for the Spatial Distribution of Plasma Components Based on Spectral Imaging. Chinese Invention Patent CN108982378B, granted 30 June 2020.

Presentations

- **Oral Presentation**: IFAC World Congress 2026, CDC 2025, AIChE 2025, AIChE 2024, BCUSC 2024, ADCHEM 2024, ESCAPE 2023, CCEC 2022, INFORMS 2022
- **Poster Presentation**: AIChE 2025, TrustML Workshop 2023, NeurIPS 2022, ICML 2022, CAC 2019

Research Experience

- **Learning Optimal Control Laws via Interpretable Machine Learning Models** — Vancouver, Canada
  Research Assistant, PhD Student *Advisor: Dr. Yankai Cao* — *Aug 2024 – May 2026*
  - Developed learning-based Model Predictive Control (MPC) laws using oblique decision trees with linear predictions at the leaf nodes.
  - Achieved control performance competitive with advanced MPC at the speed of a low-level PID controller.
  - Delivered significant online speedups: over 1,000x faster than MPC and 10x faster than neural network equivalents.
  - Extended the framework to quadcopter and multi-loop adaptive PID control, leading a five-member team.
- **Large-Scale Optimization Algorithms for Interpretable Machine Learning Models** — Vancouver, Canada
  Research Assistant, PhD Student *Advisor: Dr. Yankai Cao* — *Sep 2021 – May 2026*
  - Designed global optimization algorithms for large-scale interpretable machine learning tasks (e.g., K-Means, K-Medoids, K-Center clustering, and decision trees) with up to one billion samples, scaling 250x (K-Medoids) and 200,000x (K-Center) beyond earlier exact methods with over 20% better objectives than widely used heuristics.
  - Developed a GPU-accelerated moving-horizon method that trains deep classification trees on up to ten million samples, released as the open-source Julia package MHDETrees.jl.

- Developed reduced-space spatial branch-and-bound algorithms with customized lower bounding strategies, scenario relaxations, problem-specific bound tightening, and sample reduction techniques.
- Accelerated solution processes using hybrid CPU-GPU parallel computing to fully utilize modern hardware architectures.

- **Real-Time Fault Detection and Diagnosis for Batch Processes** — Hangzhou, China
  Research Assistant, Master's Student Advisor: Dr. Dong Ni — *Sep 2018 – Jun 2021*
  - Applied multivariate statistical analysis (e.g., PCA) and time-series models (e.g., SARIMA, LSTM) to model variable dependencies in industrial batch processes.
  - Developed a SARIMA-based drift compensation framework and a differential weighted distance-based phase alignment method, improving fault detection rates by 50% and reducing model complexity by 10-fold.
  - Designed an LSTM Encoder-Decoder architecture for real-time monitoring of nonlinear behaviors in batch processes, doubling the fault detection rate without increasing the false alarm rate.
- **Dynamic Spectral Feature Extraction for Plasma Etch Processes** — Hangzhou, China
  Research Assistant, Undergraduate Student Advisor: Dr. Dong Ni — *Oct 2017 – Jun 2018*
  - Extracted dynamic behavior using PCA and spectral peak information via wavelet decomposition.
  - Combined temporal and spectral information to derive dynamic spectral features for process monitoring.
  - Validated the method on optical emission spectra from plasma etching, enabling real-time process state detection aligned with underlying chemical mechanisms.

Work Experience

- **BC Hydro and Power Authority** — Vancouver, Canada
  Intern Research Engineer — *Aug 2023 – Apr 2024*
  - Hydro resource scheduling: applied Mesh Adaptive Direct Search to value stored water for long-term hydro planning, cutting training time from days to hours and raising modelled generation revenue by 2%.
- **Samsung (China) Research and Development Co., Ltd.** — Hangzhou, China
  Intern Software Engineer — *Apr 2017 – Sep 2017*
  - Ported the device tree seeking and reading API from the Linux kernel to U-Boot using C.

Teaching & Mentoring

- Teaching Assistant: UBC CHBE 366 Chemical Engineering Lab (Data Analysis, 200 students) — Jan 2023 – May 2025
- Mentor: Mitacs Globalink Program – Deep Learning-Based Control (2 students) — May 2023 – Aug 2025
- Mentor: Mitacs Globalink Program – Global Optimization for Clustering Models (1 student) — Jul 2024 – Oct 2024

Honors & Awards

- Wall Research Award — 2025–2027
- AIChE CAST Director's Student Presentation Award Finalist *(decision tree-based controller)* — 2025
- Westcoast Energy Inc. Jack Davis Scholarship in Energy Studies — 2024–2025
- Mitacs Accelerate Fellowship — 2023–2024
- Josephine T. Berthier Fellowship — 2023–2024
- Petrov Family Graduate Scholarship in Chemical and Biological Engineering — 2022–2023
- University of British Columbia Affiliate Fellowship — 2022–2023
- NeurIPS 2022 Scholar Award — 2022
- Best Theoretical Paper, 17th INFORMS Data Mining Workshop *(global optimal decision tree)* — 2022
- Zhejiang University Scholarship for Outstanding Students — 2018–2021

Professional Service

- **Session Organizer**: Canadian Chemical Engineering Conference 2022 – Artificial Intelligence and Machine Learning in Process Systems Engineering Session
- **Journal Reviewer**: Pattern Recognition, IEEE Transactions on Industrial Informatics, IEEE Transactions on Cybernetics, IEEE Transactions on Automation Science and Engineering, IEEE Transactions on Systems, Man, and Cybernetics: Systems, IEEE Transactions on Neural Networks and Learning Systems, Measurement, Industrial Management & Data Systems
- **Conference Reviewer**: BigData 2025, CDC 2025, ICLR 2025, BigData 2024, ACMMM 2024, CIKM 2024, ICLR 2024, CIKM 2023, NeurIPS 2023, ICML 2022
- **Others**: IET Cyber-Systems and Robotics, Blog Writer for General Audience